

Solu — See why molecules dissolve

An in-browser AI chemistry tutor that predicts a molecule's water solubility and teaches the structure-property reasoning behind it.

DSH Hacks V1 · Theme: AI × STEM Education · Web app (no install) · github.com/garyzhang1006/solu

THE PROBLEM

Chemistry students memorize the rule "like dissolves like" but rarely see why one molecule dissolves and another doesn't. Solubility depends on a tug-of-war between polarity, hydrogen bonding, and lipophilicity that stays invisible in a textbook. Meanwhile, most "AI for education" tools are thin wrappers around a chat model — non-deterministic, unverifiable, and disconnected from real scientific data.

WHAT SOLU DOES

- Type a SMILES string or pick from 14 example molecules.
- **RDKit-JS** draws the 2D structure and computes 9 descriptors live, in the browser.
- A trained **gradient-boosting model** predicts log-solubility (mol/L), converted to mg/L and a qualitative class.
- A **"Why?" panel** decomposes the prediction into per-property contributions (LogP, polar surface area, H-bond donors...) a student can learn from.
- **Quiz mode** for active recall; built-in lessons on polarity and hydrogen bonding.

THE DATA & THE MODEL

Trained on the **ESOL / Delaney** dataset — 1,128 compounds with *measured* aqueous solubility, published by Delaney (2004) and distributed via **MoleculeNet** (Wu et al., *Chem. Sci.* 2018, Stanford). Features are the interpretable Delaney descriptor family computed with RDKit.

Real academic dataset

No LLM · deterministic

100% client-side

Open source (MIT)

MODEL	RMSE	R ²
Linear (Delaney-style)	1.04	0.77
Random forest	0.80	0.87
Gradient boosting (shipped)	0.75	0.88

Held-out 20% test set. Beats the classic textbook linear equation (RMSE 1.04 → 0.75) on the same split.

WHY IT'S TECHNICALLY CREDIBLE

The model trains in Python (RDKit + scikit-learn) but runs in the browser (RDKit-JS + ONNX Runtime Web). We discovered and fixed a real train/inference skew (RDKit changed its H-bond-acceptor definition between versions), then locked correctness with automated parity gates: RDKit-JS vs RDKit-Python descriptors agree to 1e-5, and the full browser inference chain reproduces the Python prediction **bit-for-bit (diff = 0.0)**. Browser predictions are provably identical to the trained model.

TECH STACK

RDKit & RDKit-JS · scikit-learn · skl2onnx + ONNX Runtime Web · vanilla HTML/CSS/JS (no framework, no backend).

Delaney, J. S. *ESOL: Estimating Aqueous Solubility Directly from Molecular Structure*. *J. Chem. Inf. Comput. Sci.* 44, 1000–1005 (2004). · Wu, Z. et al. *MoleculeNet*. *Chem. Sci.* 9, 513–530 (2018). · Predictions are educational estimates, not laboratory values.